

CSE 220 – Signals and Linear Systems

Online 03

Topic: Continuous Fourier Transform

Duration: 30 Minutes

Background

Time-Shift Property of the Continuous Fourier Transform

The time-shift property of the Continuous Fourier Transform (CFT) states that if a signal is shifted in time as

$$y(t) = x(t - t_0), \quad (1)$$

then its Fourier Transform becomes

$$Y(f) = X(f) e^{-j2\pi f t_0}. \quad (2)$$

Problem Statement

Use the object-oriented programming (OOP) framework developed in Task 1 of the offline assignment.

Part 1: Extending the Framework

Extend the `SignalGenerator` class by implementing a new method:

`gaussian(a)`

which generates the Gaussian signal

$$x(t) = e^{-at^2}. \quad (3)$$

Note:

- The parameter a controls the width of the Gaussian signal.

Part 2: Constructing the Original Signal

1. Define the time axis as

$$t \in [-5, 5] \quad (4)$$

using at least 2000 samples.

2. Generate the signal

$$x(t) = e^{-t^2} \quad (5)$$

by setting $a = 1$.

Part 3: Time-Shifting the Signal

Let the time shift be defined as

$$t_0 = 1. \quad (6)$$

Construct the shifted signal

$$y(t) = x(t - 1). \quad (7)$$

Important: The time shift must be implemented using the OOP framework. Do *not* manually shift arrays.

Part 4: Continuous Fourier Transform

Using the `CFTAnalyzer` class:

1. Compute the Continuous Fourier Transform (CFT) of $x(t)$.
2. Compute the Continuous Fourier Transform (CFT) of $y(t)$.

Part 5: Numerical Verification

1. Plot the magnitude spectra

$$|X(f)| \quad \text{and} \quad |Y(f)|. \quad (8)$$

Use a frequency axis covering

$$f \in [-10, 10]$$

with at least 1000 samples.

2. Plot the phase spectra of both signals.
3. Verify the time-shift property by comparing:

$$|X(f)| = |Y(f)| \quad (9)$$

and

$$\angle Y(f) = \angle X(f) - 2\pi f t_0. \quad (10)$$

Part 6: Error Analysis (MSE and Phase Difference)

To numerically verify the time-shift property, compute the following error metrics.

(a) Mean Squared Error (MSE) of Magnitude

Compute the Mean Squared Error between the magnitude spectra:

$$\text{MSE}_{\text{mag}} = \frac{1}{N} \sum_{i=1}^N (|X(f_i)| - |Y(f_i)|)^2. \quad (11)$$

Print the computed value and briefly comment on whether it confirms the time-shift property.

(b) Phase Difference Error

Using the theoretical relationship

$$\angle Y(f) = \angle X(f) - 2\pi f t_0, \quad (12)$$

compute the Mean Squared Error between the measured and predicted phase:

$$\text{MSE}_{\text{phase}} = \frac{1}{N} \sum_{i=1}^N (\angle Y(f_i) - (\angle X(f_i) - 2\pi f_i t_0))^2. \quad (13)$$

Instructions

- You are **NOT** allowed to use `np.fft` or any built-in Fourier transform function.
- You **must** use `np.trapz()` or `np.trapezoid()` for numerical integration.
- Submit a single Python file named: `StudentID.py`